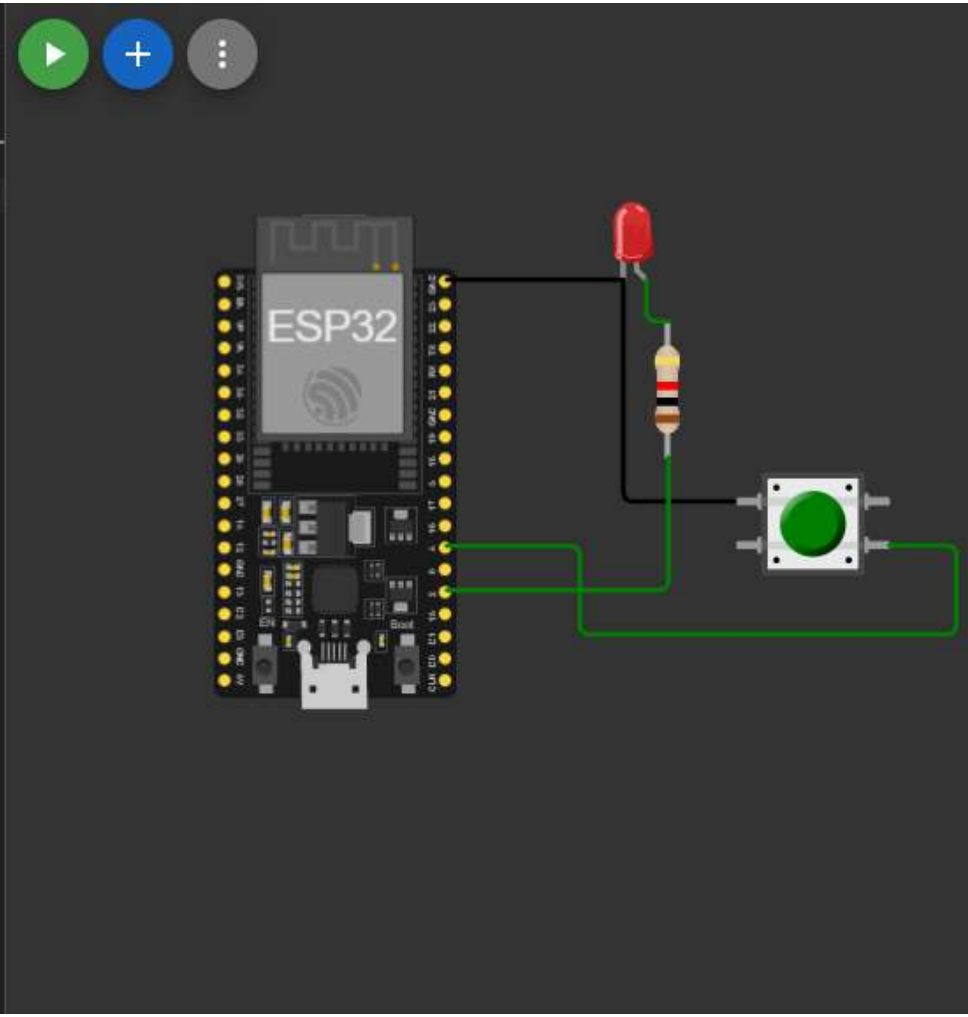```ino
#define BUTTON_PIN 4
#define LED_PIN 2
volatile bool buttonPressed = false;
void IRAM_ATTR handleInterrupt() {
buttonPressed = true; // ISR ngắn gọn, chỉ đặt cờ
}
void setup() {
Serial.begin(115200);
pinMode(BUTTON_PIN, INPUT_PULLUP);
ledcAttach(LED_PIN, 5000, 8);
attachInterrupt(digitalPinToInterrupt(BUTTON_PIN),
handleInterrupt, FALLING);
}
void loop() {
static int brightness = 0, step = 1;
if (buttonPressed) {
buttonPressed = false;
ledcWrite(LED_PIN, 0); delay(80); // nháy tắt nhanh
ledcWrite(LED_PIN, 255); delay(80); // nháy sáng nhanh
}
brightness += step;
if (brightness <= 0 || brightness >= 255) step = -step;
ledcWrite(LED_PIN, brightness);
delay(10);
}
```